Practical 8 -

701-1613-01L Landscape Patterns and Processes HS2026

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1 Interpret the landscape by eye

The visual comparison of the satellite images from 2018 and 2025 provides an opportunity to identify potential changes in the landscape of Bregaglia over a 7-year period. One of the most striking differences concerns snow cover at higher elevations. The mountain peaks located to the north of the valley show a particularly noticeable reduction in snow cover between 2018 and 2025. Assuming that both images were taken at a similar time of year, this could indicate the effects of climate change, particularly rising temperatures and the resulting reduction in snow cover. In contrast, snow cover in the southern part of the landscape appears relatively unchanged. This could partly be explained by the slightly higher elevation of this area, as well as potential differences in slope orientation and solar exposure, which may influence snow accumulation and persistence. However, differences in the timing of image acquisition and annual weather conditions could also contribute to this observation.

Another potential difference concerns vegetation colour. The slopes and smaller valleys in the upper part of the image appear to be a brighter green in 2018 than in 2025. This could reflect differences in vegetation conditions resulting from variations in precipitation and summer drought. In particular, increasingly frequent and intense dry periods associated with climate change could contribute to reduced vegetation greenness in more recent years. However, this interpretation remains speculative without additional information on the climatic conditions during the respective years.

Changes related to human activities are more difficult to identify at this scale. The spatial resolution and broad extent of the satellite images make it challenging to detect smaller-scale changes in land use or habitat structure. Moreover, urbanisation in Bregaglia is largely concentrated along the valley floor, which represents only a relatively small proportion of the overall landscape. Consequently, no major visible changes in urban development or associated habitat loss can be identified between 2018 and 2025.

Despite these observations, the interpretation of landscape changes based solely on satellite imagery has several limitations. Visual interpretation mainly reflects visible land cover and does not provide information on underlying plant composition, vegetation structure or ecological characteristics. Furthermore, comparing only two snapshots of the landscape, rather than a continuous time series, makes it difficult to distinguish long-term trends from temporary variations. Factors such as differences in the season of image acquisition, annual precipitation and drought conditions must also be considered, as they can strongly influence the appearance of vegetation and snow cover.

More generally, although landscapes are dynamic systems shaped by both natural processes and human activities, a period of only seven years may be insufficient to reveal substantial landscape-scale transformations. This is particularly relevant in Bregaglia, where steep slopes and other topographic constraints limit human development and land-use intensification.

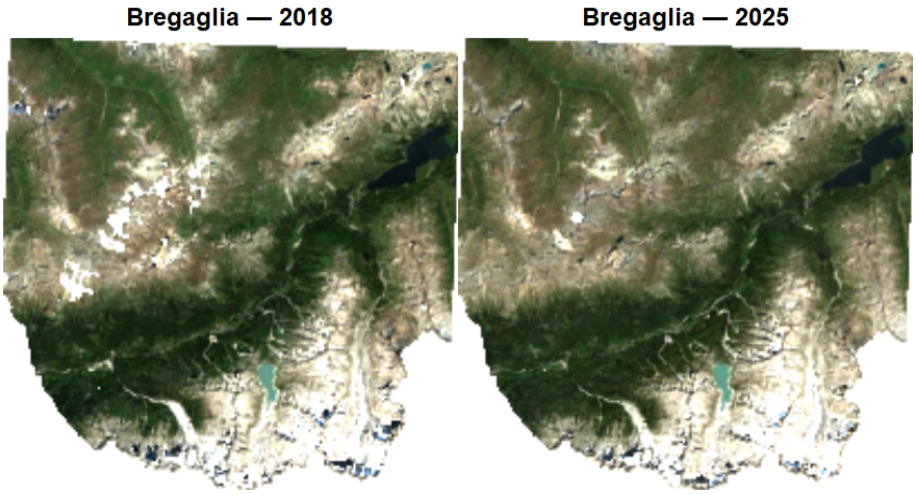


Figure 1: Comparison between Bregaglia landscape in 2018 and 2025.

2 Classify land cover automatically

Land cover in Bregaglia was classified from the 2025 Sentinel-2 composite using a Random Forest. The ten reflectance bands were complemented by five spectral indices and seven DHM25 terrain variables. Reference observations were translated to the SSP-CH legend, while ambiguous source classes and classes with fewer than 20 local observations were omitted. Observations were grouped into 5-km spatial blocks: four folds were used for model selection and a fifth fold of 367 observations was reserved for testing. This is more demanding than the tutorial's random split because neighbouring observations cannot occur in both training and test data. The final prediction was masked with the same commune boundary used for the satellite-image comparison.

Terrain produced the clearest improvement among the local experiments: balanced accuracy increased from 55.6% for the tuned spectral model to 65.9% for the tuned terrain model. A Swiss-wide hybrid was only 0.3 percentage points better, so the local model was retained as the more direct description of Bregaglia. Its detailed nine-class prediction achieved 67.8% overall accuracy and 65.9% balanced accuracy. Combining related labels after prediction into the six classes in Figure 2 raised these values to 78.2% and 76.3%, respectively. Overall accuracy is the proportion of all test observations classified correctly; balanced accuracy averages class recall and is therefore more informative for the uneven reference sample. Neither statistic directly estimates the proportion of mapped area that is correct.

Forest on the valley slopes and bare or glaciated surfaces at high elevation form the most convincing pattern when compared with the natural-colour image. Their aggregated F1 scores were 84.0%, 83.2% and 81.8%, respectively. Water was less reliable (60.5% F1), especially along narrow rivers, and isolated built-up pixels should also be treated cautiously. Sentinel-2 measures physical surface properties, whereas several reference labels also describe land use; management cannot always be inferred from one image. Moreover, a 100-m cell may contain vegetation, buildings, roads and water, and overlapping spectral signatures make such mixed pixels difficult to classify. Apparent boundaries or transitions should therefore be checked against the satellite image, aerial imagery or local field knowledge.

Table 1: Independent test performance of the principal model variants.

Model	Overall accuracy (%)	Balanced accuracy (%)	Macro-F1 (%)
Basic Bregaglia	56.7	47.1	45.7
Tuned Bregaglia spectral	61.0	55.6	55.7
Tuned Bregaglia terrain	67.8	65.9	65.2
Hybrid CH + Bregaglia terrain	**68.9**	**66.2**	**65.3**

Table 2: Class-level F1 scores after aggregation to the mapped six-class legend.

Class	Artificial	Static/bare	Forest	Grassland	Glacier/snow	Water
F1 (%)	72.4	83.2	84.0	72.1	81.8	60.5

3 Explore two possible futures

3. Two possible futures

Bregaglia is a high-alpine valley with little settlement area ($\sim$1.5 km^2 of the study area) and almost no cropland. We can therefore assume that the landscape depends largely on pasture management. To reflect this, we chose the SSP1-CH ("Efficient Switzerland") and SSP4-CH ("Unequal Switzerland") scenarios, which assume efficient land use with productive farming and a falling population with abandonment of Alpine farmland, respectively. Both scenarios share the same climate trajectory, so glacier loss is identical (-4.3 km^2; Tab. 3) and the differences between them reflect socioeconomic assumptions only.

Overall, around 21 % of the commune (74 km^2) is assigned a different class in 2080 depending on the scenario. Most of these differences lie on the middle slopes, between the settlements in the valley and the rock- and ice-covered areas above (Fig. 3). In SSP4, pastures become shrubland and open forests (-9.9 km^2 alpine pastures, $+8.6$ km^2 shrubland; Tab. 3), indicating abandonment of land. In SSP1, by contrast, shrubland is cleared back to pasture or replaced by closed forest (-18.3 km^2 shrubland, $+3.8$ km^2 pastures, $+6.1$ km^2 closed forest).

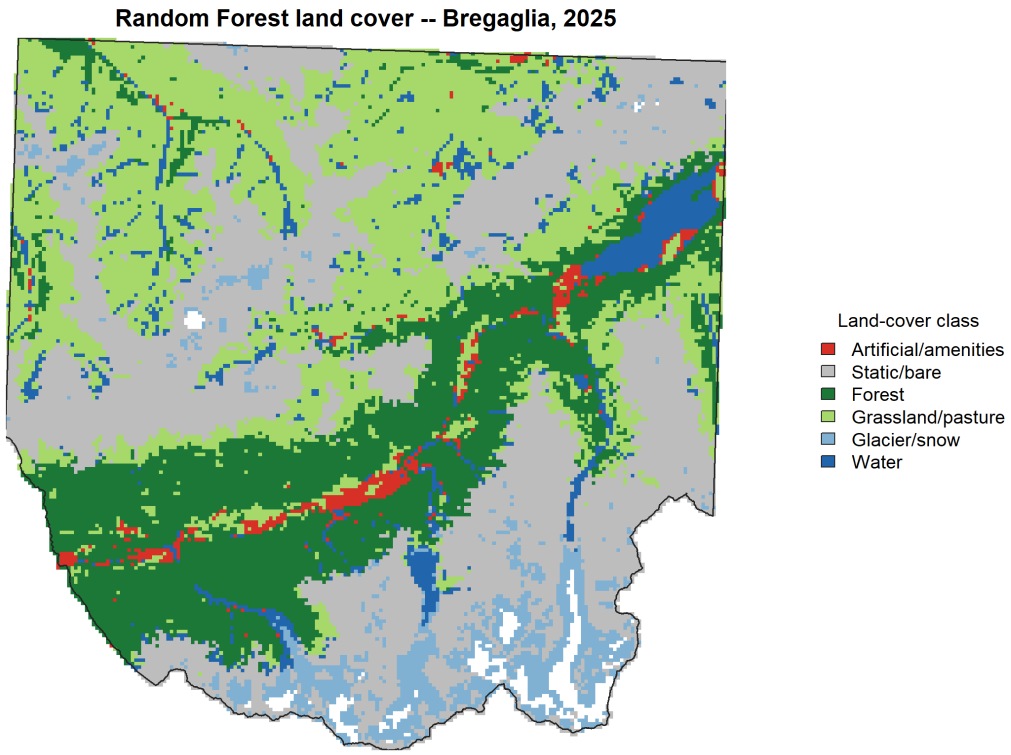


Figure 2: Random Forest land-cover prediction for Bregaglia in 2025. The fine nine-class output was aggregated after classification to the six displayed classes. Sentinel-2 bands and indices were combined with DHM25 terrain predictors at 100-m resolution; white cells have no complete predictor set and were not classified.

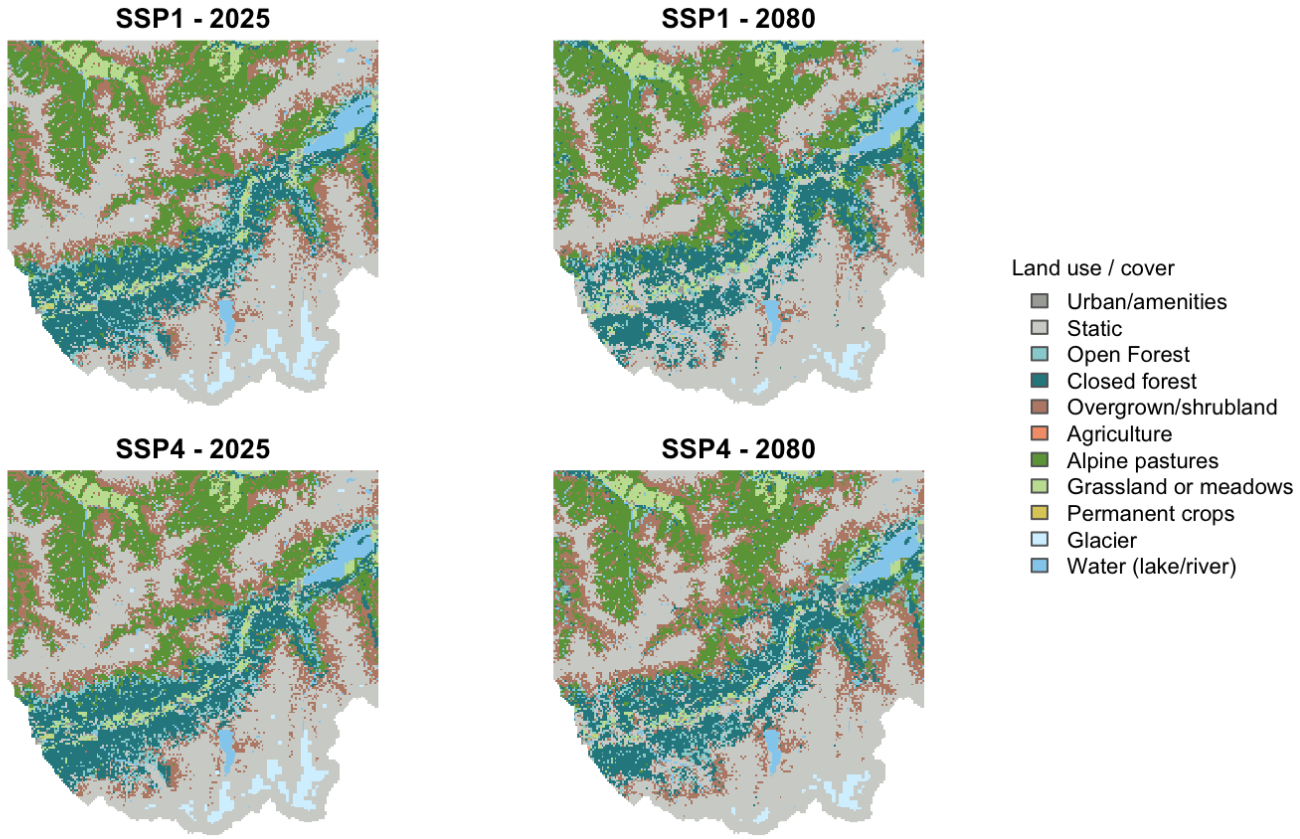


Figure 3: Projected land use in Bregaglia (100 m grid), SSP1-CH 'Efficient' and SSP4-CH 'Unequal' Switzerland, 2025 and 2080.

The steep terrain and small villages limit settlement change in both futures: urban area changes by less than 0.3 km^2 in either scenario (Tab. 3). The largest differences between the scenarios in Bregaglia concern the treeline and the pasture areas, both of which are sensitive to management.

Both scenarios also convert closed forest to *Static* (bare surfaces): 10.7 km^2 in SSP1 and 6.3 km^2 in SSP4. The scenario storylines do not explain this, so it may be a model artefact, although rockfall and debris flows could occur locally. Furthermore, the scenarios rest on national assumptions; local factors such as tourism or hydropower are not represented but could strongly influence how the landscape develops.

Table 3: Projected land-cover areas in Bregaglia in 2080 and change since 2025 ($\Delta A = A_{2080} - A_{2025}$), in km^2, for SSP1-CH and SSP4-CH (NCCS SSP-CH, 100 m grid). Commune area 345 km^2; 74 km^2 (21.5 %) differ in class between the two scenarios in 2080.

	SSP1 (Efficient)		SSP4 (Unequal)	
Class	A_{2080}	ΔA	A_{2080}	ΔA
Alpine pastures	74.4	+3.8	60.1	**−9.9**
Overgrown/shrubland	35.0	**−18.3**	62.8	**+8.6**
Closed forest	52.6	+6.1	50.4	+3.0
Open forest	15.7	+1.4	18.1	+4.5
Grassland/meadows	11.8	+0.0	10.0	−1.5
Urban/amenities	1.2	−0.2	1.6	+0.0
Glacier	2.7	−4.3	2.7	−4.3
Static	141.5	+11.5	129.1	−0.5

4 Quantify change within each scenario

Scenario 1 and 4 ...giulia